```
/*
 * Personal Diary Management System
 * Features: Login, Add, View, Search, Edit, Delete, Encryption, Change Password
 *
 * Compile: gcc diary.c -o diary
 * Run    : ./diary   (Windows e: diary.exe)
 */

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>
#include <ctype.h>

#define KEY 5                   /* encryption key (XOR) */
#define DIARY_FILE "diary.dat"   /* entry gulo ekhane thake (encrypted) */
#define TEMP_FILE  "temp.dat"    /* edit/delete er somoy temporary */
#define PASS_FILE  "pass.dat"    /* password (encrypted) */
#define MAX_TRIES  3
#define PASS_LEN   30

/* ---------- Structure: ekta diary entry ---------- */
struct Entry {
    char date[20];
    char title[50];
    char content[500];
};

/* ================================================
 *                HELPER FUNCTIONS
 * ================================================ */

void clear_screen() {
#ifdef _WIN32
    system("cls");
#else
    system("clear");
#endif
}

/* fgets diye line pore, last er '\n' bad dey. EOF hole program shesh. */
void read_line(char *buf, int size) {
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    if (fgets(buf, size, stdin) == NULL) {
        exit(0);
    }
    if (strchr(buf, '\n') == NULL) {
        /* input boro chilo, baki ongsho fele dai */
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }
    } else {
        buf[strcspn(buf, "\n")] = '\0';
    }
}

int read_int() {
    char buf[20];
    read_line(buf, sizeof(buf));
    return atoi(buf);
}

void pause_screen() {
    char buf[10];
    printf("\nEnter chaap diye ferot jan...");
    read_line(buf, sizeof(buf));
}

/* ---------- Encryption / Decryption (XOR) ----------
 * Ekbar chalale encrypt, abar chalale decrypt.
 */
void encrypt_decrypt(char *text) {
    int len = strlen(text);
    for (int i = 0; i < len; i++) {
        text[i] = text[i] ^ KEY;
    }
}

/* entry er title o content decrypt/encrypt (duitai eki kaj) */
void crypt_entry(struct Entry *e) {
    encrypt_decrypt(e->title);
    encrypt_decrypt(e->content);
}

/* boro-chhoto hater lekha upekkha kore khuje (search er jonno) */
int contains_ignore_case(const char *text, const char *word) {
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    char t[600], w[600];
    int i;
    for (i = 0; text[i] && i < 599; i++) t[i] = tolower((unsigned char)text[i]);
    t[i] = '\0';
    for (i = 0; word[i] && i < 599; i++) w[i] = tolower((unsigned char)word[i]);
    w[i] = '\0';
    return strstr(t, w) != NULL;
}

int count_entries() {
    FILE *fp = fopen(DIARY_FILE, "rb");
    struct Entry e;
    int count = 0;
    if (fp == NULL) return 0;
    while (fread(&e, sizeof(struct Entry), 1, fp) == 1) count++;
    fclose(fp);
    return count;
}

/* decrypt kora entry screen e dekhay */
void print_entry(int no, struct Entry *e) {
    printf("\n--------------------------------------\n");
    printf("#%d  |  %s\n", no, e->date);
    printf("Title  : %s\n", e->title);
    printf("Content: %s\n", e->content);
}

/* ==================================================
 *                   LOGIN
 * ================================================== */

void save_password(const char *pass) {
    char buf[PASS_LEN];
    FILE *fp;
    memset(buf, 0, sizeof(buf));
    strncpy(buf, pass, PASS_LEN - 1);
    encrypt_decrypt(buf);
    fp = fopen(PASS_FILE, "wb");
    if (fp == NULL) {
        printf("Password file likha jayni!\n");
        return;
    }
```

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    fwrite(buf, sizeof(buf), 1, fp);
    fclose(fp);
}

int load_password(char *out) {
    FILE *fp = fopen(PASS_FILE, "rb");
    if (fp == NULL) return 0;
    memset(out, 0, PASS_LEN);
    if (fread(out, PASS_LEN, 1, fp) != 1) {
        fclose(fp);
        return 0;
    }
    fclose(fp);
    out[PASS_LEN - 1] = '\0';
    encrypt_decrypt(out);
    return 1;
}

/* 1 = login hoyeche, 0 = hoyni */
int login() {
    char saved[PASS_LEN], input[PASS_LEN], again[PASS_LEN];

    if (!load_password(saved)) {
        /* prothom bar: notun password set kori */
        printf("=== Prothom bar ===\n");
        while (1) {
            printf("Notun password set korun (kompokkhe 4 okkhor): ");
            read_line(input, sizeof(input));
            if (strlen(input) < 4) {
                printf("Password chhoto!\n");
                continue;
            }
            printf("Abar likhun: ");
            read_line(again, sizeof(again));
            if (strcmp(input, again) != 0) {
                printf("Mile ni, abar chesta korun.\n");
                continue;
            }
            break;
        }
        save_password(input);
        printf("Password set hoyeche!\n");
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```
        return 1;
    }

    for (int tries = 1; tries <= MAX_TRIES; tries++) {
        printf("Password: ");
        read_line(input, sizeof(input));
        if (strcmp(input, saved) == 0) {
            printf("Login shofol!\n");
            return 1;
        }
        printf("Vul password! (%d/%d)\n", tries, MAX_TRIES);
    }
    return 0;
}

void change_password() {
    char saved[PASS_LEN], old[PASS_LEN], p1[PASS_LEN], p2[PASS_LEN];

    if (!load_password(saved)) {
        printf("Password file paoa jayni.\n");
        return;
    }
    printf("Ager password: ");
    read_line(old, sizeof(old));
    if (strcmp(old, saved) != 0) {
        printf("Vul password!\n");
        return;
    }
    printf("Notun password (kompokkhe 4 okkhor): ");
    read_line(p1, sizeof(p1));
    if (strlen(p1) < 4) {
        printf("Password chhoto!\n");
        return;
    }
    printf("Abar likhun: ");
    read_line(p2, sizeof(p2));
    if (strcmp(p1, p2) != 0) {
        printf("Mile ni.\n");
        return;
    }
    save_password(p1);
    printf("Password bodle geche!\n");
```

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}

/* ======================================================
 *                DIARY FEATURES
 * ====================================================== */

/* ---------- Add ---------- */
void add_entry() {
    struct Entry e;
    FILE *fp;

    memset(&e, 0, sizeof(e));   /* aje-baje data jeno file e na jay */

    time_t t = time(NULL);
    struct tm *info = localtime(&t);
    strftime(e.date, sizeof(e.date), "%d-%m-%Y", info);

    printf("Title: ");
    read_line(e.title, sizeof(e.title));
    if (strlen(e.title) == 0) {
        printf("Title khali rakha jabe na.\n");
        return;
    }
    printf("Content: ");
    read_line(e.content, sizeof(e.content));

    crypt_entry(&e);   /* save er age encrypt */

    fp = fopen(DIARY_FILE, "ab");
    if (fp == NULL) {
        printf("File khola jayni!\n");
        return;
    }
    fwrite(&e, sizeof(struct Entry), 1, fp);
    fclose(fp);

    printf("Entry save hoyeche (%s)\n", e.date);
}

/* ---------- View ---------- */
void view_entries() {
    struct Entry e;
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    int count = 0;
    FILE *fp = fopen(DIARY_FILE, "rb");

    if (fp == NULL) {
        printf("Kono entry nai.\n");
        return;
    }
    while (fread(&e, sizeof(struct Entry), 1, fp) == 1) {
        crypt_entry(&e);   /* decrypt */
        count++;
        print_entry(count, &e);
    }
    fclose(fp);

    if (count == 0) printf("Kono entry nai.\n");
    else printf("\nMot entry: %d\n", count);
}

/* ---------- Search ---------- */
void search_entries() {
    struct Entry e;
    char word[100];
    int mode, no = 0, found = 0;
    FILE *fp = fopen(DIARY_FILE, "rb");

    if (fp == NULL) {
        printf("Kono entry nai.\n");
        return;
    }

    printf("1. Date diye khujun (jemon 17-10-2026 ba 10-2026)\n");
    printf("2. Keyword diye khujun (title/content e)\n");
    printf("Choice: ");
    mode = read_int();
    if (mode != 1 && mode != 2) {
        printf("Vul choice!\n");
        fclose(fp);
        return;
    }
    printf("Ki khujben: ");
    read_line(word, sizeof(word));
    if (strlen(word) == 0) {
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```
        printf("Kichu likhen ni.\n");
        fclose(fp);
        return;
    }

    while (fread(&e, sizeof(struct Entry), 1, fp) == 1) {
        no++;
        crypt_entry(&e);
        int match;
        if (mode == 1)
            match = (strstr(e.date, word) != NULL);
        else
            match = contains_ignore_case(e.title, word) ||
                    contains_ignore_case(e.content, word);
        if (match) {
            print_entry(no, &e);
            found++;
        }
    }
    fclose(fp);

    if (found == 0) printf("\nKichu paoa jayni.\n");
    else printf("\n%d ta entry paoa gese.\n", found);
}

/* ---------- Edit (temp file technique) ---------- */
void edit_entry() {
    int total = count_entries();
    int n, i = 0;
    struct Entry e;
    char newtitle[50], newcontent[500];
    FILE *in, *out;

    if (total == 0) {
        printf("Kono entry nai.\n");
        return;
    }
    view_entries();
    printf("\nKon entry edit korben (1-%d, 0 = cancel): ", total);
    n = read_int();
    if (n == 0) {
        printf("Cancel kora hoyeche.\n");
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        return;
    }
    if (n < 1 || n > total) {
        printf("Vul number!\n");
        return;
    }

    in = fopen(DIARY_FILE, "rb");
    out = fopen(TEMP_FILE, "wb");
    if (in == NULL || out == NULL) {
        printf("File khola jayni!\n");
        if (in) fclose(in);
        if (out) fclose(out);
        return;
    }

    while (fread(&e, sizeof(struct Entry), 1, in) == 1) {
        i++;
        if (i == n) {
            crypt_entry(&e);   /* decrypt */
            printf("\nNotun title (Enter dile ager ta thakbe)\n[%s]: ", e.title);
            read_line(newtitle, sizeof(newtitle));
            printf("Notun content (Enter dile ager ta thakbe)\n[%s]: ", e.content);
            read_line(newcontent, sizeof(newcontent));

            struct Entry ne;
            memset(&ne, 0, sizeof(ne));
            strcpy(ne.date, e.date);
            strcpy(ne.title, strlen(newtitle) ? newtitle : e.title);
            strcpy(ne.content, strlen(newcontent) ? newcontent : e.content);
            crypt_entry(&ne);  /* abar encrypt */
            e = ne;
        }
        fwrite(&e, sizeof(struct Entry), 1, out);
    }
    fclose(in);
    fclose(out);

    remove(DIARY_FILE);
    rename(TEMP_FILE, DIARY_FILE);
    printf("\nEntry update hoyeche!\n");
}
```

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/* ---------- Delete (temp file technique) ---------- */
void delete_entry() {
    int total = count_entries();
    int n, i = 0;
    char ans[10];
    struct Entry e;
    FILE *in, *out;

    if (total == 0) {
        printf("Kono entry nai.\n");
        return;
    }
    view_entries();
    printf("\nKon entry delete korben (1-%d, 0 = cancel): ", total);
    n = read_int();
    if (n == 0) {
        printf("Cancel kora hoyeche.\n");
        return;
    }
    if (n < 1 || n > total) {
        printf("Vul number!\n");
        return;
    }

    printf("Entry #%d sotti delete korben? (y/n): ", n);
    read_line(ans, sizeof(ans));
    if (ans[0] != 'y' && ans[0] != 'Y') {
        printf("Delete kora hoyni.\n");
        return;
    }

    in = fopen(DIARY_FILE, "rb");
    out = fopen(TEMP_FILE, "wb");
    if (in == NULL || out == NULL) {
        printf("File khola jayni!\n");
        if (in) fclose(in);
        if (out) fclose(out);
        return;
    }

    while (fread(&e, sizeof(struct Entry), 1, in) == 1) {
```

```
        i++;
        if (i != n) {   /* oi entry bade baki shob copy */
            fwrite(&e, sizeof(struct Entry), 1, out);
        }
    }
    fclose(in);
    fclose(out);

    remove(DIARY_FILE);
    rename(TEMP_FILE, DIARY_FILE);
    printf("Entry delete hoyeche!\n");
}

/* ======================================================
 *                    MAIN
 * ====================================================== */

int main() {
    int choice;

    if (!login()) {
        printf("\nOnek bar vul password. Program bondho.\n");
        return 1;
    }
    pause_screen();

    while (1) {
        clear_screen();
        printf("=========== MY DIARY ===========\n");
        printf("1. Add Entry\n");
        printf("2. View All Entries\n");
        printf("3. Search Entry\n");
        printf("4. Edit Entry\n");
        printf("5. Delete Entry\n");
        printf("6. Change Password\n");
        printf("0. Exit\n");
        printf("================================\n");
        printf("Choice: ");
        choice = read_int();
        printf("\n");

        switch (choice) {
```

```
            case 1: add_entry();       break;
            case 2: view_entries();    break;
            case 3: search_entries();  break;
            case 4: edit_entry();      break;
            case 5: delete_entry();    break;
            case 6: change_password(); break;
            case 0:
                printf("Bye! Bhalo thakben.\n");
                return 0;
            default:
                printf("Vul choice!\n");
        }
        pause_screen();
    }
}
```